

Benchmark RADAR: Living Search Engine for Retrieval and Discovery of AI Benchmark Research

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Abstract

Benchmark researchers and model developers need to find relevant evaluations, trace their sources, and determine which reported scores can be compared. Benchmark Radar collects benchmark papers, repositories, datasets, catalog entries, and model reports in a searchable system. It records which benchmarks model reports mention and preserves the evaluation settings attached to numeric scores. As of 2026-09-06, the collection contained 10,702 source observations, meaning records collected from public sources, and 6,304 artifacts after observations with matching identifiers were grouped. Its catalog held 1,283 searchable benchmark records. Model reports mentioned 110 benchmarks and supplied 322 scores across 86 benchmark tracks, each collecting results for a named benchmark. The web dashboard, RSS feed, JSON downloads, and local search clients provide access to this evidence. We describe the collection and search methods, examine source coverage and benchmark adoption, and assess the limits of score comparisons when evaluation settings differ.

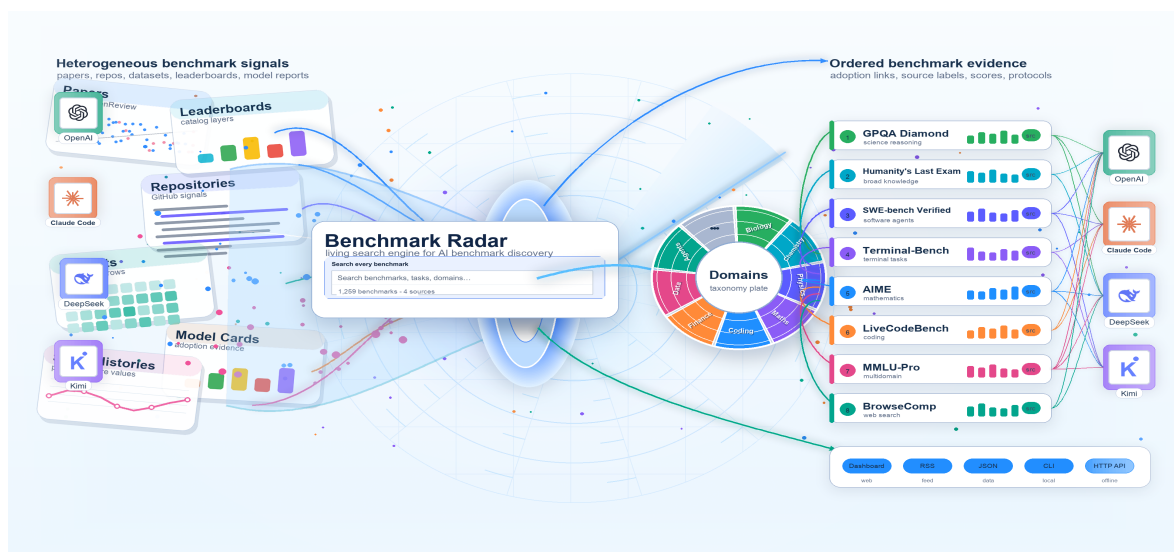


Figure 1: Benchmark Radar collects papers, repositories, datasets, and model reports, organizes them by topic, and makes them searchable alongside benchmark mentions and reported scores.

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1 Introduction

The Transformer architecture established attention-based sequence modeling as a foundation for modern large language models (LLMs) [51]. Model families such as GPT-4 and Claude Sonnet span changing capabilities and deployment settings [2, 36]. Comparing these models requires benchmarks that expose failures and distinguish capability gains from changes in prompts, data splits, or evaluation procedures [15, 39, 42, 45].

Evaluation also matters beyond general-purpose chat. In recommender systems, Transformer-style sequence models and LLM-based methods support user, item, and preference modeling, with efficient training needed at production scale [22, 43, 49]. Foundation models are also used to model biological, geoscientific, and physical processes [8, 44, 50, 56, 61]. In social and political analysis, language models support text annotation and simulated human samples, while network methods quantify balance and polarization dynamics [3, 10, 26, 28, 35, 46]. Benchmark scores inform claims of progress across these scientific, industrial, and decision-support settings.

Benchmarks differ in the abilities they test. Broad evaluations of knowledge and reasoning include MMLU [15], GPQA [42], and Humanity’s Last Exam [39, 62]. Others isolate more specific capabilities: SWE-bench measures whether models can resolve real GitHub issues [20], LiveCodeBench tests code generation with date-based splits to reduce contamination from training data [19], and Terminal-Bench and Long-Horizon-Terminal-Bench test command-line agents on operational tasks [29, 32]. Domain-specific evaluations include FinanceBench for finance and economics [18], MATH for mathematics [16], SciBench for scientific problem solving [53, 57, 59], LAB-Bench for biology [25], and ChemBench for chemistry [33]. Their tasks and scoring rules specify which model behaviors count as successful performance.

Finding a benchmark, its dataset or repository, and reports of its use requires searching paper servers, code hosting sites, dataset hubs, vendor releases, blogs, and benchmark catalogs. Existing resources such as LLM Stats, OpenCompass, and Artificial Analysis provide leaderboards, evaluation platforms, and model-analysis views [4, 30, 37]. To our knowledge, no existing system centers on benchmark search while combining daily discovery, catalog records, mentions in model reports, and scores with documented evaluation settings and sources.

Benchmark Radar gathers benchmark-related artifacts from public sources, groups observations with matching identifiers, and exposes a searchable index with source labels, dates, benchmark mentions, and scores. We present the search system, describe the collection and ranking methods needed to reproduce its outputs, and examine benchmark coverage, adoption in model reports, source concentration, and gaps in evaluation settings.¹

2 Methodology

2.1 System Overview

Benchmark Radar adapts the daily collection approach of BuilderPulse, which answers a fixed question set from more than ten sources each morning [58]. It applies this approach to benchmark discovery, catalog search, and score histories. Readers can look for recent releases, search for a task or benchmark name, inspect mentions in model reports, and compare scores collected under matching evaluation settings. Table 1 lists these uses and the interfaces that support them.

The source systems expose different fields and identifiers. Benchmark Radar converts their records into common fields for names, descriptions, source labels, identifiers, dates, and supporting documents.

¹Source code: <https://github.com/ktwu01/benchmark-radar>.

Interface	Reader question	Role in Benchmark Radar
Today	What appeared or changed?	Ranks new records and links the source evidence.
Search box	Which records match this name, task, or domain?	Matches names, phrases, and query words in the benchmark index.
Leaderboard	Which benchmarks do labs report?	Counts benchmark mentions across curated model-card documents.
Scores	Which reported scores can be compared?	Groups scores by benchmark version, metric, and evaluation settings.
Trends and map	Which topics and sources recur?	Records source coverage so changes in collection can be considered when comparing topics.
CLI and HTTP	Can an analyst reproduce search locally?	Use one query service and a shared JSON response format.

Table 1: Reader questions and the interfaces used to answer them.

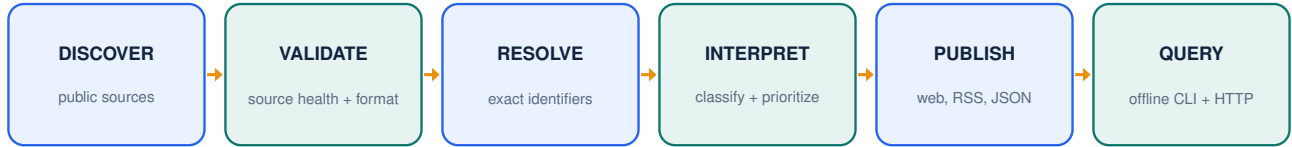
A source observation is a collected record; an artifact is a paper, repository, dataset, release, or page identified from those observations. The benchmark catalog retains one record per benchmark within each source, so similarly named entries from different sources remain distinct. Model reports describe model development and evaluation, and their benchmark mentions provide evidence of adoption [34]. Here, adoption means a documented mention; it does not imply a comparable numeric score.

2.2 Collection and Source Health

Connectors are programs that retrieve records from public sources. Each collection run searches a 48-hour window, records counts and errors by source, removes future-dated rows, and requires healthy responses from a configured set of core sources before publication. The core sources at the cutoff were arXiv [5], Hugging Face Hub [17], and GitHub Search [12]. Optional routes include GitHub organizations, Hugging Face Papers, Kaggle datasets [21], Zenodo [60], OpenReview [38], Semantic Scholar [24], GitHub Releases [11], OpenAlex [40], Brave Search [6], 24 institution-operated research or engineering feeds, and a separate collector of public attention through Hacker News search on Algolia [1]. All three core sources were healthy at the cutoff. Semantic Scholar was rate-limited, Brave Search lacked an API key, and OpenReview returned no eligible rows. Appendix A lists the healthy connectors and monitored feeds.

The collection run at the cutoff fetched 867 rows, reduced them to 837 candidate records after removing duplicates, classified 322 as eligible, and recommended 110 for readers to review. Two collections ran within the cutoff window; after merging, the daily page contained 322 records, including 110 recommendations. Published records retain the source URL, retrieval time, parser version, source identifier, timestamps, authors or organizations when supplied, and a SHA-256 checksum for verifying the stored record. Raw responses and credentials are excluded.

Dated snapshots, record formats, parser versions, and checksums allow the published data to be rebuilt and checked. Publication requires healthy core sources; failures in optional sources remain visible. Installed clients validate an update before activating it. Identity resolution relies on exact identifiers and reviewed links between source records. The evaluated release passed 1,236 tests after a clean rebuild. Section 4 describes the remaining limitations.



Dated source records and fixed collection settings make the results reproducible.

Figure 2: Collection and publication stages. Connectors retrieve public records and check source health and record formats. Matching DOI, arXiv, OpenReview, GitHub, or Hugging Face identifiers link observations to artifacts. Classification and priority scoring organize the records before publication. Web pages, feeds, and local clients expose the collected evidence.

2.3 Identity, Ranking, and Search

The resolver groups observations when they share a DOI, arXiv identifier, OpenReview identifier, GitHub repository, or Hugging Face artifact. Similar titles alone do not force a merge; they remain separate until review confirms that they describe the same artifact. This policy reduces false joins but leaves possible duplicates visible when papers, repositories, and catalogs lack a common identifier.

Search covers four catalog sources, as summarized in Table 2: LLM Stats, OpenCompass Hub, Artificial Analysis, and Model reports. The index contains 1,283 source records, including 110 model-report benchmarks. Of these model-report benchmarks, 86 have reported scores and 24 have no scores; both groups remain searchable. Search matches query words and phrases against record fields, including benchmark names. It searches all four sources together and preserves each result’s source label. The same query against the same index produces the same ranking.

A separate priority score for daily recommendations combines relevance (35%), evidence (20%), recency (20%), and adoption (25%) on a 0–100 scale. These weights describe daily reading priority; search relevance depends on the query. The scoring configuration is included with the published data.

Catalog source	Rows	Primary use
LLM Stats	687	Benchmark names and descriptions with source provenance.
OpenCompass Hub	461	Paper, repository, release, and dataset links with source-specific identity fields.
Artificial Analysis	25	Current commercial evaluation catalog entries.
Model reports	110	Benchmarks mentioned in model reports, including records without scores.
Total	1,283	All four sources, retaining each record’s source label.

Table 2: Catalog sources in the Benchmark Radar query index.

2.4 Adoption and Score Curation

Benchmark Radar records benchmark mentions and numeric scores from model-card and system-description documents [34]. This collection includes 37 model documents from 12 organizations. They mention 110 benchmarks whose names have been reconciled within the model-report collection. When a document contains an interpretable numeric value and sufficient context, the score archive records the benchmark version and metric (the instrument), along with evaluation settings (the protocol). These settings include examples supplied in the prompt (shots), tool access, attempts, and the evaluator when available. The archive contains 322 scores across 86 benchmark tracks. A score track groups

results for a named benchmark; comparisons within it still require matching instruments and protocols. Scores from benchmark registries remain outside these model-report score histories unless they can be connected to a comparable protocol [9, 34].

3 Results

3.1 Coverage at the Cutoff

At the 2026-09-06 cutoff, the cumulative discovery collection held 10,702 source observations and 6,304 artifacts grouped by exact identifiers. Primary or structured sources supplied 10,473 observations, or 97.86% of the collection. This category covers scholarly records, institution-operated feeds, code repositories, and dataset or artifact hosts. Discovery observations, benchmark records, model-report mentions, and scores count different objects and must not be added together. Table 3 defines each unit.

10,702 source observations across 45 snapshots	6,304 artifacts grouped by matching identifiers	1,283 searchable benchmark records, including unscored	4 catalog sources with source labels retained
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An observation is a collected source record. Matching identifiers link observations to artifacts. The catalog counts benchmark records separately for each source.

Data cutoff: 2026-09-06. Catalog records and discovery observations are separate units.

Figure 3: Source observations, resolved artifacts, searchable benchmark records, and catalog sources as of 2026-09-06.

Unit	Count	Interpretation
Snapshot	45	Dated collection records, including 41 observed and 4 simulated snapshots.
Source observation	10,702	Collected records with their source and retrieval information.
Unique artifact	6,304	Papers, repositories, datasets, releases, or pages grouped by matching identifiers.
Searchable entry	1,283	One benchmark record per source, including records without scores.
Adoption benchmark	110	Benchmarks mentioned in model reports, with names reconciled within that collection.
Model-card document	37	Curated reports from 12 organizations.
Curated score	322	Numeric results across 86 benchmark tracks.

Table 3: What each count measures at 2026-09-06.

3.2 Source Composition and Reliability

Five source labels account for 9,532 of 10,702 observations, or 89.1% of the cumulative collection. Because these sources supply most observations, their retrieval caps, rate limits, and availability can

change the apparent topic mix. At the cutoff, the required sources were healthy, while optional routes varied in yield because of rate limits, missing credentials, or empty result windows.

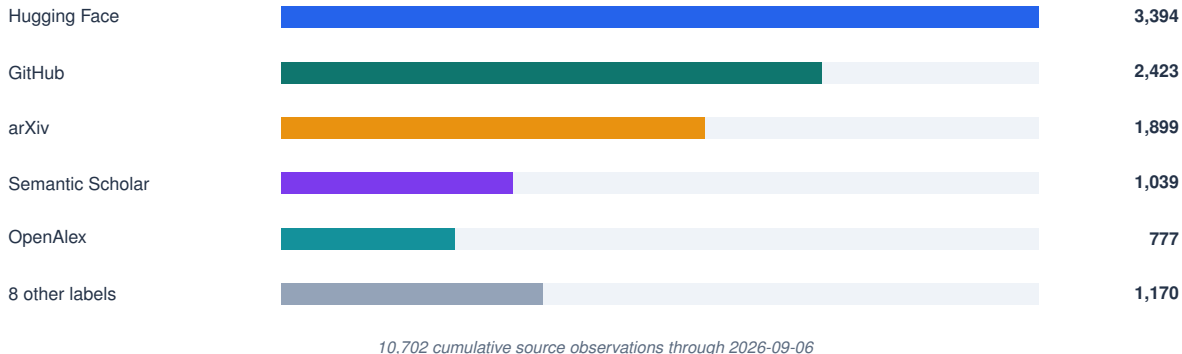


Figure 4: Discovery observations by source through 2026-09-06. The bars count collected observations, including repeat sightings, rather than distinct benchmarks.

Fifty-nine of 6,304 artifacts were linked across more than one source, or 0.94% of the artifact set. Matching exact identifiers reduces false joins; links between papers, code, and datasets still require review when identifiers are missing or inconsistent.

3.3 Relationship to Existing Catalogs

Benchmark Radar imports records from three benchmark registries: LLM Stats, OpenCompass Hub, and Artificial Analysis. These projects have different goals. LLM Stats publishes live benchmark rankings and benchmark-family pages [30]; OpenCompass is primarily an evaluation platform with a searchable dataset-statistics page [37]; and Artificial Analysis publishes model evaluations and a documented intelligence-benchmarking methodology [4].

The index contains 687 records from LLM Stats, 461 from OpenCompass Hub, 25 from Artificial Analysis, and 110 from Model reports, yielding 1,283 searchable source records. Records without scores remain in this total; source membership does not establish a trust ranking. Direct comparisons of totals would be misleading because record definitions, update frequency, source coverage, and evaluation settings differ across systems. Benchmark Radar combines these catalog records with daily discovery, artifact links based on matching identifiers, mentions in model reports, and scores with documented evaluation settings.

3.4 Benchmark Adoption in Model Reports

In the model-report collection, eight benchmarks appeared in documents from at least six organizations: GPQA Diamond [42], Humanity’s Last Exam [39], SWE-bench Verified [20], Terminal-Bench [32], AIME [31], LiveCodeBench [19], MMLU-Pro [54], and BrowseComp [55]. GPQA Diamond appeared in 27 of 37 documents from 11 organizations, making it the most commonly mentioned benchmark in these reports. This concentration suggests that frontier-model reporting has partly converged on a common set of evaluations, even as researchers introduce benchmarks in other areas.

A benchmark can be widely mentioned without providing comparable scores across dates. Comparing scores can be misleading when reasoning budget, tool access, attempts, evaluator, or date-based test split differs [9, 34]. In the snapshot, eight bounded metrics had a raw best score within five points of their ceiling: AIME [31], Arena-Hard [27], DeepSearchQA [13], HMMT [14], MATH-500 [16], Math-

Vision [52], SWE-bench Verified [20], and tau2-bench [47]. Headroom is the gap between a reported score and the metric’s upper bound. Appendix B separates these raw best scores from results with matching evaluation settings across dates. Six benchmarks gained model-card mentions after their last interpretable score, so newer adoption evidence did not always come with newer comparable scores.

4 Limitations and Future Work

Lexical search can miss paraphrased benchmark names, renamed tasks, or descriptions of related ideas. Optional semantic retrieval could find such records by meaning, while retaining the source, field, and word matches that let readers inspect each result [41]. Source coverage also depends on public endpoints, API credentials, and rate limits. Counts need units and a data cutoff so readers can distinguish changes in collection from changes in benchmark activity.

Exact identifiers reduce false joins but leave possible links between papers, code, and datasets unresolved when identifiers are absent. Further identity review should link repositories to papers and document the evidence for each decision. In the snapshot, 12,594 scores from benchmark registries remain outside the model-report score histories because they lack protocol fields such as prompt examples, tool access, attempts, evaluator, date-based test splits, or evaluation software. Recording these settings would allow more scores to be compared.

Task-level coverage is also incomplete. The capability-classification component, KW-Bench, has 5,661 tracks without complete task-capability labels at the cutoff. Here, a classification track is an artifact or a declared task within a benchmark suite, drawn from discovery observations; it is a different unit from a catalog record or a model-report score track. These missing labels limit conclusions about which abilities the collected tasks test. Completing them would help readers identify the capability a task requires. Counts in the dashboard and local outputs also need consistent labels for observations, artifacts, benchmark records, mentions, and scores.

5 Conclusion

Benchmark Radar brings benchmark discovery, catalog search, model-report mentions, and score histories into one system while retaining the sources behind each record. The evaluated snapshot combines 37 public source routes, a 1,283-record search index, and adoption evidence from 37 model documents.

Readers can use this evidence to find candidate benchmarks and inspect their reporting history. Comparing scores still requires checking the benchmark version and evaluation settings. Missing identifiers leave some artifact relationships unresolved, and missing protocol details prevent many scores from supporting comparisons across dates.

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Appendix

A Source Inventory and Health

Table 4 lists source connectors that completed successfully at the cutoff. Row counts are measured before duplicate removal and ranking. A connector can be healthy with zero eligible rows if its request succeeds and the response can be parsed. Core sources cover papers, shared model or dataset artifacts, and code; all must be healthy before publication. Failures in optional sources are reported but do not block publication.

Connector	Role	Cutoff status	Core
arXiv	Primary papers via selected AI, language, vision, and software categories	Healthy; 0 rows	Yes
Hugging Face Hub	Datasets and Spaces connected to benchmark artifacts	Healthy; 92 rows	Yes
GitHub Search	Code repositories and benchmark artifacts	Healthy; 300 rows; at cap	Yes
GitHub Organizations	Reviewed organization repositories	Healthy; 4 rows	No
Hugging Face Papers	Community-surfaced papers	Healthy; 0 rows	No
Kaggle Datasets	Public benchmark datasets	Healthy; 32 rows	No
Zenodo	DOI-bearing research artifacts	Healthy; 57 rows	No
OpenReview	Conference submissions	Healthy; no eligible rows	No
GitHub Releases	Curated first-party releases	Healthy; 6 rows	No
OpenAlex	Scholarly discovery metadata	Healthy; 200 rows	No

Table 4: Healthy direct discovery connectors at the current cutoff.

Table 5 lists research and engineering feeds operated by the institutions being monitored. They can surface benchmark papers, datasets, model announcements, and engineering posts. The feed collector yielded one relevant record at the cutoff. Some healthy feeds produced zero matches after relevance filtering. Searches restricted to an organization’s website cover institutions that lack a verified feed.

Feeds 1-12	Feeds 13-24
Meituan Engineering	Ai2
OpenAI News	Together AI
Google AI	Sakana AI
Google DeepMind	Qwen
Google Research	Ollama
Apple Machine Learning Research	Stability AI
AWS Machine Learning	Nomic AI
Hugging Face Blog	Replicate
Microsoft Research	NVIDIA Developer
NVIDIA AI Blog	IBM Research
Mistral AI	Databricks
Meta Research	LangChain

Table 5: First-party research and engineering feeds monitored by Benchmark Radar.

B Protocol-Controlled Saturation Audit

The eight metrics identified in Section 3 have raw best scores within five points of their upper bound. This appendix examines whether the archive also contains scores obtained under matching evaluation settings on at least two dates.

Every metric in this appendix has a percentage scale with an upper bound of 100. *Raw headroom* is 100 minus the highest reported score, regardless of setup. *Repeat-controlled headroom* is 100 minus the highest score among setups observed on at least two dates with identical instruments and protocols. A setup specifies the benchmark version, metric, and evaluation settings. Differences in attempts, tool access, reasoning budget, judge, or date-based test split keep scores in separate groups. Two dates support a paired comparison within that setup; they do not establish a trend across the field.

All eight raw best readings come from a setup that appears on a single date. Four of the eight benchmarks have a different setup repeated across dates, and each of those repeated setups is a two-date, single-organization pair. The remaining four have no repeated setup at all, so their headroom cannot be assessed under repeat control.

Benchmark	Raw best	Raw gap	Repeat gap	Repeated setup	Repeated gap ≤ 5 ?
AIME	99.2	0.8	20.2	2 dates, 1 org	No
Arena-Hard	95.6	4.4	7.7	2 dates, 1 org	No
DeepSearchQA	95	5	n/a	none	Unknown
HMMT	96.9	3.1	4.8	2 dates, 1 org	Yes
MATH-500	98	2	n/a	none	Unknown
MathVision	97.8	2.2	n/a	none	Unknown
SWE-bench Verified	96	4	19.4	2 dates, 1 org	No
tau2-bench	99.3	0.7	n/a	none	Unknown

Table 6: Raw and repeat-controlled headroom, in percentage points. Each repeated setup spans two dates from one organization. Only HMMT has a repeated-setup gap within five points; four benchmarks have no repeated setup. All raw-best setups occur on a single date.

Table 7 counts how many raw readings and repeated setups fall within each headroom threshold. Only four benchmarks have a repeated setup, so the other four are unknown and excluded from that denominator.

Threshold	Raw readings	Repeated setups
≤ 5	8 / 8	1 / 4 (4 unknown)
≤ 3	4 / 8	0 / 4 (4 unknown)
≤ 2	3 / 8	0 / 4 (4 unknown)

Table 7: Benchmarks within each headroom threshold, measured in percentage points. Fractions use the eight raw readings or the four benchmarks with repeated setups as their denominators.

None of the setups that produced the raw best scores was observed on a second date. Table 8 gives the evaluation settings and the evidence available for each benchmark.

Benchmark	Score, evaluation setup, and evidence limits
AIME	AIME’s 99.2 raw best leaves 0.8 points of headroom under the AIME 2026, temp 1.0, top_p 0.95, max gen 163840 tokens, GPT-5.5 (medium) judge setup, but that setup appears on one date. The closest repeated setup leaves 20.2 points, outside the five-point threshold.
Arena-Hard	Arena-Hard’s 95.6 raw best leaves 4.4 points of headroom under the thinking mode setup, but that setup appears on one date. The closest repeated setup leaves 7.7 points, outside the five-point threshold.
DeepSearchQA	DeepSearchQA’s 95 result leaves 5 points of headroom under the F1, reasoning=max setup. No setup spans two dates, so the archive supports only this protocol-specific reading.
HMMT	HMMT’s raw best is an isolated 96.9 result. A separate think max, pass@1 setup spans 2 dates from 1 organization and leaves 4.8 points of headroom. This supports a protocol-specific paired comparison.
MATH-500	MATH-500’s 98 result leaves 2 points of headroom under the thinking mode setup. No setup spans two dates, so the archive supports only this protocol-specific reading.
MathVision	MathVision’s 97.8 result leaves 2.2 points of headroom under the reasoning=max, with Python setup. No setup spans two dates, so the archive supports only this protocol-specific reading.
SWE-bench Verified	SWE-bench Verified’s 96 raw best leaves 4 points of headroom under the adaptive thinking, max effort, averaged over 5 trials setup, but that setup appears on one date. The closest repeated setup leaves 19.4 points, outside the five-point threshold.
tau2-bench	tau2-bench’s 99.3 result leaves 0.7 points of headroom under the Thinking (High), Telecom setup. No setup spans two dates, so the archive supports only this protocol-specific reading.

Table 8: Evaluation settings and limits of the near-ceiling results. A small gap applies to the stated setup and does not establish that the benchmark is saturated across settings.

The audit groups reported scores by instrument and protocol, checks the number of distinct dates in each group, and calculates the gaps summarized above. The reproduction command and score archive are listed in Appendix E.

C Worked Use Case: Prior-Art Check for a New Evaluation

This case describes one contributor’s use of Benchmark Radar to check related work before designing an evaluation. It is a single workflow, with no measured timing or accuracy outcome.

The contributor wanted to determine whether a proposed evaluation duplicated existing work. The session ran during August 2026 and surveyed recent work on credit assignment in agentic training,

using small Qwen-series models as a requirement for reproducible baseline experiments. Comparing a candidate design against existing work normally requires manual searches, whose completeness is difficult to establish.

The contributor gave the task to a coding agent together with the public setup instructions for Benchmark Radar. The agent installed the command-line client and the Benchmark Radar Skill, a set of instructions for using the client. It then downloaded the local corpus and queried candidate records offline.

```

what can i do with benchmark-radar cli?

+ Thought: 328ms

$ benchmark-radar --help
usage: benchmark-radar [-h] [--author-output AUTHOR_OUTPUT]
                        [--author-contacts AUTHOR_CONTACTS]
                        [--author-repo-limit AUTHOR_REPO_LIMIT]
                        [--author-contributors AUTHOR_CONTRIBUTORS]
                        [--author-emails] [--config CONFIG] [--output OUTPUT]
                        [--json-output JSON_OUTPUT]
                        [--social-output SOCIAL_OUTPUT] [--items ITEMS]
                        [--repo REPO] [--channels CHANNELS]
                        [--existing-body EXISTING_BODY] [--since SINCE]
                        [--until UNTIL] [--snapshot-dir SNAPSHOT_DIR]

Click to expand

+ Thought: 531ms

The CLI has two groups:

Consumer (query) commands – for searching and inspecting local benchmark data:
- search "query" --scope catalog|radar[all] --json – discover benchmarks (catalog – verified records, radar – recent unverified evidence like papers/repos/datasets)
- show "key-or-slug" --json – inspect one record in detail
- recent --json – newest Radar evidence
- status --json – local data health/provenance
- init / sync – set up or refresh the local data
- serve --host 127.0.0.1 --port 8765 – local HTTP interface for the same queries

Maintainer commands – for operating the radar pipeline (not for ordinary use): run, rebuild, backfill, migrate, rescore, simulate-history, export, classify, authors, social, normalize-external, build-data-release.

For your local benchmark search use case, you'll mostly need search, show, recent, status, and serve.

Build · DeepSeek V4 Flash DeepSeek
..... esc Interrupt 158.1K (15%) · $8.83 ctrl+p commands

```

Figure 5: A coding agent installs the Benchmark Radar client and its usage instructions, then runs local queries.

The agent inspected which paper, repository, and dataset links were recorded for each candidate. Figures 6 and 7 show records with a repository link and no dataset link. The paper-link field differs between them. A missing link describes the collected record; it does not establish that the artifact has not been released.

```
"rank": 18,
"recommended": false,
"score": 18.75,
"slug": null,
"snapshot_date": "2026-09-01",
"source": "Hugging Face Papers",
"source_id": "2608.28476",
"updated_at": "2026-08-28T00:00:00+00:00",
"url": "https://huggingface.co/papers/2608.28476"
},
{
  "categories": [
    "benchmark"
  ],
  "description": "LTABC-KM is a long-tail-aware adaptive artificial bee colony-K-means for unsupervised image clustering. It leverages density-aware coresets and graph-propagation assignment to recover minority tail modes without labels, achieving significant Macro-F1, NMI and ARI improvements on imbalanced benchmarks with favorable quality-computation trade-offs.",
  "has_dataset": false,
  "has_paper": false,
  "has_repo": true,
  "has_size": false,
  "key": "radar:github:Lutra11/LTABC-KM",
  "kind": "radar",
  "languages": [],
  "match": {
    "idf_coverage": 0.4391,
    "matched_fields": [
      "description"
    ],
    "matched_tokens": [
      "assignment"
    ],
    "missing_tokens": [
      "credit",
      "training"
    ],
    "phrase_fields": [],
    "query_coverage": 0.3333,
    "ranking_algorithm": "bm25f_v3",
    "reason": "1 of 3 unique query tokens matched across fields; IDF coverage 0.44",
  }
}
```

Build · DeepSeek V4 Flash DeepSeek

D:\projects\radar-test187.3K (19%) · \$0.04ctrl+p commands

Figure 6: A retrieved record with a repository link, no dataset link, and a false paper-link field. These fields describe the links recorded by the client.

```
"source": "GitHub",
"source_id": "mdkamranalam/credixa",
"updated_at": null,
"url": "https://github.com/mdkamranalam/credixa"
},
{
  "categories": [
    "benchmark"
  ],
  "description": "Long-horizon agentic tasks require large language models (LLMs) to iteratively retrieve, integrate, and maintain dispersed information across multi-turn interactions, but preserving all interaction histories leads to a continuously growing working context. Recent proactive context management methods allow models to edit their own working context with specialized tools, yet they still face three key limitations: (1) a limited toolset restricted to search, deletion, and summarization, with no support for global planning, long-term memory, and adaptive compression; (2) inefficient exploration that treats context management actions uniformly despite their heterogeneous impacts on final outcomes; and (3) coarse-grained credit assignment that assigns the final trajectory-level reward to all intermediate context editing actions during RL. To bridge these gaps, we introduce ContextPilot, a proactive context management framework for long-horizon agentic reasoning. Our approach systematically augments the toolset with planning, long-term memory, and soft context offloading tools. We further propose an RL method tailored for context management, which uses context and entropy variation to identify critical editing decisions for branch sampling and estimates action-level advantages from all branched trajectories that pass through the corresponding context editing action. Experiments on long-context QA and deep search tasks show that ContextPilot achieves stronger performance with a more compact working context, consistently outperforming existing baselines across various base models and benchmarks. Code is available at https://github.com/Tencent/ContextPilot.",
  "has_dataset": false,
  "has_paper": true,
  "has_repo": true,
  "has_size": false,
  "key": "radar:hugging face papers:2608.28476",
  "kind": "radar",
  "languages": [],
  "match": {
    "idf_coverage": 0.8392,
    "matched_fields": [
      "description"
    ],
    "matched_tokens": [
      "assignment",
      "credit"
    ]
  },
}
```

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D:\projects\radar-test187.3K (19%) · \$0.04ctrl+p commands

Figure 7: A second retrieved record with paper and repository links but no dataset link. The missing dataset link does not establish its release status.

Benchmark Radar search matches words and phrases, so the agent also ran its own web search and cross-checked the two candidate sets before accepting a record. The additional search checked for related work described in different terms. The agent still had to judge whether each retrieved record was relevant to the proposed evaluation.

```

"kind": "radar",
"languages": [],
...
Click to expand

$ benchmark-radar search "credit assignment training" --scope radar --json

...
"query_coverage": 0.3333,
"ranking_algorithm": "bm25f_v3",
"reason": "1 of 3 unique query tokens matched across fields; IDF coverage 0.44",
"retrieval_score": 10.824384,
"score_components": {
  "bm25f": 10.824384,
  "name_bonus": 0.0,
  "phrase_bonus": 0.0
},
"modality": null,
"name": "deepvisionaiofficial/Automation_AI-Assignment-Checker",
"openness": null,
"published_at": "2026-08-10T05:56:17+00:00",
"publisher": "",
"rank": 6,
"recommended": false,
"score": 36.32,
"slug": null,
"snapshot_date": "2026-08-10",
"source": "GitHub",
"source_id": "deepvisionaiofficial/Automation_AI-Assignment-Checker",
"updated_at": null,
"url": "https://github.com/deepvisionaiofficial/Automation_AI-Assignment-Checker"
},
{
  "categories": [
    "benchmark",
    "agentic"
  ]
}

```

use internal rubrics to compute retrieval score

Build · DeepSeek V4 Flash DeepSeek

D:\projects\vradar-test 187.3K (10%) · \$0.04 ctrl+ip commands

Figure 8: A candidate record with query-word matches and retrieval-score components for the agent to inspect when judging relevance.

The session ended with a focused summary table of related work that the contributor judged complete enough to act on.

#	Work	Published	Qwen base model(s)	Credit-assignment focus	Agentic benchmarks used	Link
1	SRPO: Self-Reflective Policy Optimization for Long-Horizon Reasoning	2026-08-25	Qwen3-8B (also Qwen3-1.7B / 32B scaling)	Reflection-conditioned dense token-level signals convert sparse terminal reward into training signal	AIME'24, WebShop, ALFWorld, SWE-Bench-Lite	https://arxiv.org/abs/2608.23493 · https://github.com/Galleons2029/SRPO
2	ContextPilot: Teaching Agents for Proactive Context Management via Fine-grained RL	2026-08-28	Qwen3-8B, Qwen3-14B	Coarse-grained → action-level credit assignment over branched trajectories during agentic RL	InfBench (InfiniteBench), NovelQA, LongMemEval, BrowseComp+	https://arxiv.org/abs/2608.28476 · https://github.com/Tencent/ContextPilot
3	SkillGate: Training In-Policy Skill Selection in Long-Horizon Agents	2026-08-21	Qwen3.5-9B (SFT checkpoint, RL init)	"Selector credit starvation": partitions token support into two credit channels (outcome vs. action-local advantage)	5 agentic benchmarks incl. SkillsBench, SWE (16-candidate skill slate)	https://arxiv.org/abs/2608.18852
4	CIPO: Contextual Information Policy Optimization for Search Agents	2026-08-06	Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct	Dense turn-level credit (EALR) to evidence-using reasoning actions, combined with outcome reward	HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle + 3 OOD	https://arxiv.org/abs/2608.06128
5	MoRSE: Task-Oriented Multi-Agent System with Mixture of Role-Subtask Experts	2026-08-10	Qwen3-4B-Instruct (one of three backbones; also Llama-3.1-8B, Gemma-4-31B)	Hierarchical GRPO with two-layer credit assignment (isolates expert vs. routing quality)	Code-generation benchmarks (multi-agent), held-out task domains	https://arxiv.org/abs/2608.09251

Figure 9: Summary table of recent work on credit assignment in agentic training, assembled during the session.

Figure 10 shows a comparison table from an earlier benchmark-design effort by one contributor. It covers ten related benchmarks across six design dimensions, with each cell read from a source paper. The contributor reported that assembling it took more effort than any part of that project except producing the benchmark data. Benchmark Radar retrieves candidates for this kind of comparison and exposes the matching words and fields for inspection; the comparison itself requires reading the source evidence.

Bench Name	End-to-End Tasks	Fine-Grained Eval	Researcher-Quality Eval	Data Generation	Multi-Harness Eval	#Tasks
MLE-Bench (Chan et al. 2025)	✗	✓	✗	Transfer&Compose	✓	75
MLGym-Bench (Nathani et al. 2025)	✗	✓	✗	Automatic	✗	13
EXP-Bench (Kon et al. 2025)	✓	✗	✗	Automatic	✓	461
ResearchCodeBench (Hua et al. 2026)	✗	✓	✗	Transfer&Compose	✗	212
MLR-Bench (Chen et al. 2026)	✓	✗	✗	Automatic	✓	201
PaperBench (Starace et al. 2025)	✗	✓	✗	Transfer&Compose	✗	8316
AstaBench (Bragg et al. 2025)	✓	✓	✗	Transfer&Compose	✓	2400+
InnovatorBench (Wu et al. 2025)	✓	✗	✗	Transfer&Compose	✗	20
AIRS-Bench (Lupidi et al. 2026)	✗	✓	✗	Automatic	✓	20
COMPOSITE-Stem (Waters et al. 2026)	✓	✓	✗	Manual	✗	70
ScienceBoard (Sun et al. 2025)	✗	✓	✗	Manual	✗	169

Figure 10: A prior-art comparison table assembled manually for an earlier benchmark-design effort. Each row is one related benchmark and each column one design dimension, all read from the source papers by hand.

Query wording can change the candidate set because search matches words rather than meaning. The session combined Benchmark Radar with the agent’s web search, so it does not isolate either tool’s contribution. Repository and dataset availability can also change after retrieval. The summary table and screenshots document one workflow, without a controlled comparison. Additional session evidence is available in the project’s public [issue #492](#).

D Contributor Credit

Junjie Zhou prepared the protocol-controlled saturation audit in Appendix B, its machine-readable table, and the corresponding report revision, documented in [issue #457](#). Jiayu Wang contributed the worked use case in Appendix C and its supporting evidence, documented in [issue #492](#).

E Reproducibility, Access, and Citation

This paper evaluates Benchmark Radar v0.10.0 at Git commit 76a6180 with a data cutoff of 2026-09-06. The data build combines dated collection snapshots, benchmark registry records, and model reports, then classifies tracks and packages the results with checksums. Installed clients validate the checksum and record format before activating an update. The evaluation used a clean data rebuild with linting, formatting checks, catalog normalization, classification, release construction, and the full test suite; all 1,236 tests passed. The manuscript is authored in LaTeX, separately from the data generators. The project provides a DOI and a Citation File Format record for citation [7, 23, 48].

Artifact	Canonical or permanent location
Archived paper (v0.9.0)	https://doi.org/10.5281/zenodo.22167102
Source code	https://github.com/ktwu01/benchmark-radar
Dashboard	https://benchmark-radar.org/
Discovery observations (JSON)	https://benchmark-radar.org/data/radar.json
Benchmark catalog	https://benchmark-radar.org/data/benchmark-index.json
RSS	https://benchmark-radar.org/feed.xml
Citation metadata	https://github.com/ktwu01/benchmark-radar/blob/main/CITATION.cff

Table 9: Access and citation artifacts for Benchmark Radar.

The reported counts were computed from the cumulative discovery data, benchmark catalog, model registry, model-report and score archives, normalized catalog records, and collection configuration. Their repository locations are `site/data/radar.json`, `site/data/benchmark-index.json`, `site/data/models.json`, `data/model_cards.yml`, `data/benchmark_scores.yml`, `data/catalog/`, and `config.yml`. The dated numerical inputs used by the manuscript and figures are stored in `figure-data.tex` beside the LaTeX source.

To reproduce the data build from a clean checkout, install the project and run `benchmark-radar normalize-catalog`, `benchmark-radar classify`, and `benchmark-radar build-data-release`, in that order. The command `python -m benchmark_radar.saturation_audit` regenerates the protocol audit from the score and model-report archives, including the score identifiers, evaluation groups, and exclusions behind the appendix tables. Run `make` in `docs/technical-report/latex/` to build this manuscript from its dated numerical inputs. Rebuilding the data at a later revision does not update those inputs automatically.

The DOI in Table 9 identifies the archived v0.9.0 paper. This manuscript describes v0.10.0 at the cutoff above. The dashboard continues to change; citations of its live counts need a retrieval date.